

VARAD VISHWARUPE

PhD Researcher · Human-Centered AI · Alignment & AI Safety

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RESEARCH INTERESTS

Human-Centered AI · Alignment · AI Safety · Human–AI Interaction · Alignment Evaluation · Ethics of AI · Philosophy

EDUCATION

DPhil (PhD), Computer Science — University of Oxford, UK Oct 2024 – Sep 2027

Inaugural Computer Science Scholar, Oxford Institute for Ethics in AI

Supervisors: **Sir Nigel Shadbolt, Prof. Marina Jirotko, Prof. Ivan Flechais** ·

Assessed coursework grade: 88/100 ·

M.Tech, Artificial Intelligence & Machine Learning — BITS Pilani, India Sep 2022 – Aug 2024

GPA 9.81/10 · Ranked 3rd of 122 · Bronze Medal · Best Dissertation Prize

MSc, Advanced Computer Science — University of Oxford, UK Oct 2019 – Sep 2020

Distinction (78.6%) · Department Merit Scholarship

B.Eng., Information Technology — MIT College of Engineering, Pune, India Aug 2012 – Aug 2016

Distinction (79.67%) · Departmental Rank 1st of 178 · Gold Medal

RESEARCH & INDUSTRY EXPERIENCE

Student Researcher Intern, Alignment Team — Google DeepMind Jul 2025 – Sep 2025

Semantic stress-testing of RLHF-trained agentic models

London, UK

- Developed semantic stress-testing methods evaluating robustness of RLHF-trained models under paraphrase and inquiry-phrasing variation; characterised how small lexical shifts produce divergent reasoning paths in agentic settings.
- Contributed to alignment evaluation work on stability and value-preserving behaviour, examining how optimisation objectives shape implicit normative responses in deployed configurations.

Graduate Teaching Assistant — University of Oxford

Oct 2024 – Present

Department of Computer Science

Oxford, UK

- Teaching Assistant for Machine Learning, Probabilistic Inference (Neural Networks), Computer Architecture, Computer Vision, and Quantum & Probabilistic Computing.
- Mentored 8 undergraduate and MSc students across Oxford colleges; interviewed 18 candidates for undergraduate Computer Science admissions at Keble College; marked the Mathematics Admissions Test (MAT).

Research Scientist II, Alexa AI — Amazon

Oct 2020 – Aug 2022

Automatic Speech Recognition & Voice Services

Hyderabad, India / London, UK

- Led core ASR and wake-word detection research within Alexa Voice Services and Alexa Skills Kit; production work delivered a 12.7% increase in user satisfaction and 8% reduction in ASR error rates serving millions of users.
- Managed a six-person team of engineers and data scientists; shipped few-shot learning pipelines on Amazon SageMaker and AWS Lambda; awarded Impactful Team Player recognition.

Software Development Engineer I, Azure Machine Learning — Microsoft

Oct 2017 – Sep 2019

ML-as-a-service infrastructure and Azure Stack HCI

Hyderabad, India

- Built ML-as-a-service features within Azure Machine Learning Studio enabling scalable model deployment and integration of core and advanced ML models for enterprise users.
- Contributed to hybridisation of on-premises HCI infrastructure with Azure cloud services; the deployment expanded Azure Stack HCI adoption by approximately 6%.

Machine Learning Intern — IBM Research

Jun 2016 – Aug 2016

DB2 / Bluemix proprietary technologies

New Delhi, India

- Carried out data anonymisation on large-scale time-varying consumer datasets hosted on IBM Cloud; improved anonymisation efficiency for TTL solutions from 71% to 88%.

SELECTED PUBLICATIONS

Accepted

- **Vishwarupe, V.**, et al. (2026). *From Dual-Use Awareness to Dual-Use Agency: Epistemic Distance and the Structural Limits of Ethical Responsibility in AI Research*. ICLR 2026.
- **Vishwarupe, V.**, Flechais, I., Jirotko, M., Shadbolt, N. (2026). *“To LLM, or Not to LLM”: How Designers and Developers Navigate LLMs as Tools or Teammates*. CHI 2026, Barcelona, Spain. ACM. doi:10.1145/3772363.3798953
- **Vishwarupe, V.**, Jirotko, M., Shadbolt, N., Flechais, I. (2026). *The Collaboration Gap in Human–AI Work: Grounding and Repair Conditions for Stable Collaboration*. ECSCW 2026, Germany.

- **Vishwarupe, V.**, Flechais, I., Jirotko, M., Shadbolt, N. (2026). *From Rights to Rites: Expectations Management in Smart-Home AI*. HCI International (HCII) 2026, Montreal, Canada. Springer.
- **Vishwarupe, V.**, et al. (2026). *Pluralism as a Safety Property: Auditing RLHF-Tuned LLMs in Labour Contexts*. ACM FAccT 2026.
- **Vishwarupe, V.**, Kuklani, R. et al. (2025). Semantic Jailbreaks and RLHF Limitations in LLMs: A Taxonomy, Failure Trace, and Mitigation Strategy. *International Journal of Computer Applications*, 187, 187–196.

Under Review / In Submission

- **Vishwarupe, V.**, Flechais, I., Jirotko, M., Shadbolt, N. (2026). *NeurIPS Should Require Reproducibility Standards for Frontier AI Safety Claims*. NeurIPS 2026 Position Paper Track
- **Vishwarupe, V.**, et al. (2026). *Position: Deployment-Relevant Alignment Cannot Be Inferred from Model-* audit of 16 LLM alignment benchmarks (Cohen's $\kappa = 0.87$).
- **Vishwarupe, V.**, et al. (2026). *The Dual-Use Agency Gap: Missing Actors in Technical AI Governance Access Architectures*. ICML 2026
- **Vishwarupe, V.**, et al. (2026). *Beyond the Artifact Layer: A Framing Note on the Locus of Governance for AI-Assisted Science*. ICML 2026

Published Earlier

- **Vishwarupe, V.** et al. (2025). Breaking the Black Box: Securing and Auditing Edge-Deployed LLMs via Shard Traceability. *International Journal of Computer Applications*, 187(27), 44–49.
- **Vishwarupe, V.**, Joshi, P. et al. (2022). Explainable AI and Interpretable Machine Learning: A Case Study in Perspective. *ACM ISCSI 2022, Portugal*. [Best Paper Award].
- **Vishwarupe, V.**, Bedekar, M., Pande, M. et al. (2022). Bringing Humans at the Epicentre of Artificial Intelligence: A Confluence of AI, HCI, and Human-Centred Computing. *Procedia Computer Science*, Elsevier.

SELECTED RESEARCH PROJECTS

Virtuous Intelligence Framework (VIF) : Designed a deployment-time alignment framework operationalising three mechanisms — scoping, signalling, and repair — derived from grounded-theory analysis of human–agent interaction breakdowns. Currently at v3.0 with prototype Live VIF Monitor wired to a frontier LLM API for controlled validation.

Repair-Oriented Alignment Evaluation: Structured dual-coded audit (Cohen's $\kappa = 0.87$) of 16 leading LLM alignment benchmarks against an eight-dimension interactional rubric. Findings: verification support is absent across the ecosystem; process steerability is nearly absent; the interactional tier is fragmented. NeurIPS 2026 Position-track submission.

Reproducibility Standards for Frontier AI Safety Claims: Three-tier disclosure framework (T1 public / T2 controlled / T3 restricted) with claim inventory, mandatory scope statements, and graduated sanctions, modelled on ACM artefact-review badging. NeurIPS 2026 Position-track submission.

DeepMind Alignment Summer Project: Semantic stress-testing of RLHF-trained agentic models — evaluating robustness under paraphrase and inquiry-phrasing variation, and how optimisation objectives shape implicit normative responses.

SKILLS

AI Safety & Alignment: Alignment evaluation design, RLHF, Constitutional AI principles, red-teaming, jailbreak analysis, semantic stress-testing, repair-oriented benchmarks, scalable oversight, ARENA programme and BlueDot Impact courses.

Programming: Python (fluent, production-grade), C++, R, SQL, JavaScript (React, Node.js), HTML/CSS.

ML Frameworks & Infrastructure: PyTorch, TensorFlow, JAX, Haiku, scikit-learn, Hugging Face, Pandas, OpenCV, AWS (EC2, Lambda, SageMaker), Azure ML.

Empirical & HCI Methods: Constructivist grounded theory, semi-structured interview design, dual-coded thematic analysis, mixed-methods study design, RCT design, NVivo.

HONOURS & SERVICE

- Inaugural Computer Science Scholar, Oxford Institute for Ethics in AI — first CS researcher in the Institute's history.
- Funded PhD admits from Oxford, Cambridge, UCL, and Harvard (acceptance rates < 10%).
- Top 20 Young Scientists (out of 835 applicants) — Global Partnership on AI (GPAI) Summit 2023, New Delhi; presented research on AI to the Hon. Prime Minister of India.
- Young Researcher, ACM Heidelberg Laureate Forum 2023 (200 selected globally).
- Best Paper Awards: ACM ISCSI 2022 (Portugal); ACM CIIS 2022 (China); Springer ICT4SD 2018 (London).
- Conference Reviewer: NeurIPS, ICML, ACM SIGCHI, ECCV, SIGIR, ACM FAccT, ICLR (2022–2026).
- Peer Reviewer: WIREs Data Mining and Knowledge Discovery (Wiley); International Journal of Intelligent Systems (UK).
- India's Top 30 under 30: Excellence in Technology Awardee 2023.

GRANTED PATENTS

- AI-driven intelligent payment alerts (ZA202208837B);
- Intelligent hybrid Twitter spam classifier (ZA202108302B);
- Content-relevance and priority-filtered text-message notifications (IN201621024128).